OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	Salem Generating Station
SYSTEM:	Emergency Operating Procedures
TASK:	Perform Actions for a Loss of All AC Power (Energize a 4KV Vital Bus from a Station Power Transformer)
TASK NUMBER:	1150140501
JPM NUMBER:	15-01 NRC Sim g
ALTERNATE PATH:	☐

K/A NUMBER: EPE 055 EA1.07

IMPORTANCE FACTOR: 4.3 (RO) 4.5 (SRO)

APPLICABILITY:
EO ☐ **RO** ☒ **STA** ☐ **SRO** ☒

EVALUATION SETTING/METHOD: Simulator / Perform

REFERENCES: S2.OP-SO.DG-0001, Rev. 39, S2.OP-SO.4KV-0001, Rev. 32

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 10 minutes

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By:	**G Gauding** **Instructor**	**Date: 4-18-16**
Validated By:	**Hancock / Russell** **SME or Instructor**	**Date: 8-16-16**
Approved By:	**Training Department**	**Date:** 10/11/16
Approved By:	**Operations Representative**	**Date:** 10/11/16

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY:

GRADE: ☐ **SAT** ☐ **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

NAME: ______________________

DATE: ______________________

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions for a Loss of All AC Power (Energize a 4KV Vital Bus from a Station Power Transformer)

TASK NUMBER: 1150140501

Simulator Setup: IC-256 (with 24ASD racked down and control power off)

INITIAL CONDITIONS:

1. Unit 2 is recovering from a loss of all AC Power.
2. The crew is performing step 16 of 2-EOP-LOPA-1.
3. 23 AFW pump is supplying all required AFW flow.
4. 2A EDG was started first and is providing power to 2A Vital Bus
5. 2B EDG was just started and is powering 2B 4KV vital bus.
6. 2C 4KV Vital Bus remains deenergized and unavailable.
7. 24 Station Power Transformer (SPT) has just been energized.
8. With 24 SPT now available, the CRS has directed that S2.OP-SO.DG-0001 Section 5.8 and S2.OP-SO.4KV-0001 Section 5.3.6 be utilized to swap 2A Vital Bus from 2A EDG to 24 SPT.

INITIATING CUE:
You are the Control Board Operator. Complete Section 5.8 of S2.OP-SO.DG-0001, and then utilize Section 5.3.6 of S2.OP-SO.4KV-0001 to restore power to 2A Vital Bus from 24 SPT. Prerequisites and Precautions and Limitations have been reviewed and signed.

Successful Completion Criteria

1. **All critical steps completed.**
2. **All sequential steps completed in order.**
3. **All time-critical steps completed within allotted time.**
4. **JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made**

Task Standard for Successful Completion:

- Energize 2A 4KV vital bus using 24 SPT

NAME: ______________________

DATE: ______________________

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions for a Loss of All AC Power (Energize a 4KV Vital Bus from a Station Power Transformer)

	STEP No.	STEP (* denotes critical task)	STANDARD	EVAL S/U	COMMENTS
		Provide marked up copies of S2.OP-SO.DG-0001 and S2.OP-SO.4KV-0001.			
	SO.DG-1 5.8.1	IF 2A EDG is paralleled on 2A 4KV Vital Bus….	Determines 2A Diesel Generator is not paralleled on 2A 4KV Vital Bus.		
 * *	5.8.2	IF 2A Vital Bus is isolated and energized from 2A EDG, THEN: • **PLACE** redundant equipment in service as necessary to support deenergizing 2A Vital Bus • **STOP** all 2A Vital Bus loads using Attachment 1	Places redundant equipment in service as needed: • 22 CCW pump **Note:** 1 SW pump with the TGA isolated is sufficient for maintaining SW header pressure Stops the following equipment (if running): • **21 CCW pump** • **21 SW pump** • 21 CFCU • 21 Aux Building Exhaust Fan • 21 SWGR Room Supply Fan • 21 SWGR Room Exhaust Fan • 21 BAT pump • 21 FHB Exhaust Fan • **21 AFW pump** • 21 SI / 21 CS pumps		

	STEP No.	STEP (* denotes critical task)	STANDARD	EVAL S/U	COMMENTS
*	5.8.3	**OPEN** 2A DIESEL GENERATOR BREAKER by performing one of the following: • **OPEN** 2AD1AX6D, 2A DIESEL GENERATOR BREAKER (64' Swgr Rm), or • **PRESS** 2A BREAKER OPEN pushbutton (2CC3), or • **PLACE** 2A-DF-GCP-3, GENERATOR CIRCUIT (BCS), to TRIP (2A D/G Control Panel).	Opens 2A DIESEL GENERATOR BREAKER by performing one of the following: • Directs opening of 2AD1AX6D, 2A DIESEL GENERATOR BREAKER (64' Swgr Rm), or • Presses 2A BREAKER OPEN pushbutton (2CC3), or • Directs placing 2A-DF-GCP-3, GENERATOR CIRCUIT (BCS), to TRIP (2A D/G Control Panel).		
	5.8.4	**ALLOW** Diesel to run unloaded for ≥ 3 minutes prior to stopping the EDG.	**Cue: 2A EDG will be shutdown by local operator.**		
	5.8.5	If diesel unloading…	Determines step is N/A		
	SO.4KV-1 5.3.6	IF 2A 4KV Vital Bus is to be energized from 24 SPT,THEN:			
	5.3.6.A	Direct NEO to **RACK UP** 2AD1AX24ASD, 24 STATION POWER TRANSFORMER INFEED BREAKER (64' Swgr Rm).	Contacts NEO and directs them to rack up 2AD1AX24ASD, 24 STATION POWER TRANSFORMER INFEED BREAKER. **Simulator Operator**: **MODIFY REMOTE ED21D to ON,** then report 2AD1AX24ASD, 24 STATION POWER TRANSFORMER INFEED BREAKER is racked up.		
*	5.3.6.B	**PRESS** Mimic Bus 2A VITAL INFEED 24ASD pushbutton AND **ENSURE** Console Bezel 24ASD MIMIC BUS INTLK CLOSE SELECTION illuminates.	Depresses Mimic Bus 2A VITAL INFEED 24ASD pushbutton AND determines the Console Bezel 24ASD MIMIC BUS INTLK CLOSE SELECTION illuminates.		

*	5.3.6.C	**PERFORM** the following: 1. **PRESS** AND **HOLD** control console 24ASD CLOSE pushbutton. 2. **RELEASE** pushbutton when 24ASD indicates CLOSED. 3. **ENSURE** the following: a. Console bezel 24ASD MIMIC BUS INTLK CLOSE SELECTION is extinguished. b. 2A 4KV Vital Bus voltage is 4.275 - 4.336KV (normal) OR 4.330 - 4.417KV (single source of off-site power). c. OHA J-17, 2A 4KV VTL BUS UNDRVOLT, is clear.	Presses and Holds control console 24ASD CLOSE pushbutton, then releases pushbutton when 24ASD indicates closed. Checks Console bezel 24ASD MIMIC BUS INTLK CLOSE SELECTION is extinguished. Determines 2A Vital Bus voltage is 4.275 - 4.385KV (single source of off-site power. Determines OHA J-17 2A 4KV VTL BUS UNDRVOLT is clear.		
			Terminate JPM.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

✓ ✓ J 1. Task description and number, JPM description and number are identified.

✓ ✓ J 2. Knowledge and Abilities (K/A) references are included.

✓ ✓ J 3. Performance location specified. (in-plant, control room, or simulator)

✓ ✓ J 4. Initial setup conditions are identified.

✓ ✓ J 5. Initiating and terminating Cues are properly identified.

✓ ✓ J 6. Task standards identified and verified by SME review.

✓ J 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

✓ J 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. ~~29~~ 32 Date 8/16/16

J 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.

J 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

J 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: J A R Hancock Date: 8/16/16

SME/Instructor: W. Russell Russell Date: 8/16/16

SME/Instructor: ______________ Date: ______

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